

LESSON 7-4 • SESSION 1 • PRACTICE • FORM B

Comparing Numbers on a Number Line

Learning target – I can compare two rational numbers using their positions on a number line. • 6.NS.C.7a

REMEMBER

- On a number line, the number **farther right is greater**. Any positive number is greater than any negative number.
- For negatives, the one **closer to 0** is greater. Bigger digits do not mean a bigger negative number: $-13 > -40$.
- For decimals, use the same number of digits: $-0.06 > -0.60$.

WORKED EXAMPLE

Compare -14 degrees and -5 degrees.**Step 1** Plot -14 and -5 on a number line.**Step 2** -5 is to the right of -14 .**Step 3** Write the comparison: $-14 < -5$.**Answer** -5 is greater; -14 is colder**WORDS** **greater than** (*mayor que*) • **less than** (*menor que*) • **inequality** (*desigualdad*)

PRACTICE

show your work • label units

1. Write
- $<$
- or
- $>$
- in each blank.

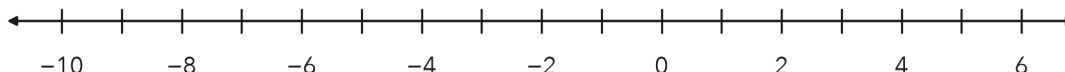
a. -6 ____ -11

b. -3.75 ____ 2.25

c. -9 ____ -1

d. 0 ____ -7

- 2.
- Trivia Scores.**
- Scores: Maya
- -8
- , Eli
- 4
- , Nora
- -3
- , Caleb
- 0
- .

Hint The score farthest right is greatest.

- a. Plot and label all four scores.
- b. Compare Nora and Maya using $>$. _____
- c. Greatest score ____ Least score ____

- 3.
- Two Accounts.**
- One account ended at
- -28
- dollars. Another ended at
- -53
- dollars. Which balance is greater?

Hint Which is closer to 0?

Show your work.

Greater _____

- 4.
- Freezer Check.**
- One freezer holds at
- -21
- degrees. Another holds at
- 3
- degrees. Which reading is colder?

Show your work.

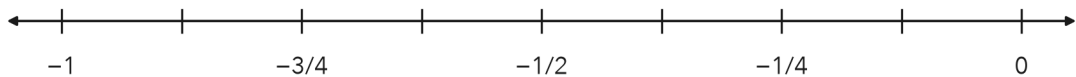
Colder _____

- 5.
- Two Cave Floors.**
- Cave A has a floor at
- -13
- m. Cave B has a floor at
- -31
- m. Write a comparison using
- $<$
- or
- $>$
- . Which cave is deeper?

Hint The deeper floor is farther left.

Deeper _____

6. **Eighths.** Plot $-\frac{5}{8}$ and $-\frac{3}{8}$. Which number is greater?



Greater _____

7. **Kofi's Claim.** Kofi says $-0.7 > -0.07$ because seven tenths is more than seven hundredths. Is he right? Which number is greater?

Hint Write -0.7 as -0.70 .

Show your work.

Greater _____

8. **Three Accounts.** Three accounts hold -60 , 15 , and -9 dollars.

- a. Order the balances from least to greatest. _____
- b. Which balance has the largest absolute value? _____
- c. Why is that balance not the greatest? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Owen's Work.** Owen wrote $-22 > -7$. He said 22 comes after 7 when you count, so -22 must be greater. Explain his mistake. Then write the correct comparison.

TEST PRACTICE

choose the best answer

10. Which symbol makes $0 \underline{\hspace{1cm}} -4$ true?
A $<$ **B** $>$ **C** $=$ **D** none of these
11. Which number is greatest?
A -12 **B** -2 **C** -20 **D** -5

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

$-\frac{5}{8}$ Z	-28 U	$-\frac{3}{8}$ E	-60 S	$-3 > -8$ N	-21 M	-53 Q	Cave B B	-0.7 X	-0.07 R
#2b		#3	#4	#5	#6	#7	#8b		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____